Impact of a 10% Oil Price Shock on Lesotho

Results from the Lesotho Quarterly Projection Model (QPM)

Model: Lesotho QPM v2 (Recalibrated) • Based on IMF CR/2023/269
Method: Perfect foresight simulation with iterative expectations
Date: March 2026

Executive Summary

This report presents the results of a **10 percent oil price shock** simulated through the Lesotho Quarterly Projection Model. The shock is modeled as a one-time 10 percentage point increase in oil price inflation, which propagates through the economy via commodity cost channels unique to Lesotho's CPI structure.

Key findings:

- Lesotho inflation rises by **0.32 pp on impact**, driven by Lesotho's higher oil weight in CPI (8%) relative to South Africa (5%).
- The output gap expands by **0.19 pp** (peaking at quarter 5), as higher inflation lowers the real interest rate under the peg, creating an accommodative monetary stance that the CBL cannot independently offset.
- The real exchange rate appreciates (-0.08 pp on impact), improving Lesotho's external competitiveness metric while reflecting higher domestic prices.
- Foreign exchange reserves improve modestly as the REER appreciation reduces interventionist pressure, lowering the risk premium by up to **0.18 pp**.
- The shock is **persistent** (oil price AR coefficient = 0.70), with inflation effects lingering for approximately 8–10 quarters.

1. The Oil Price Shock

We model a **one-time, 10 percentage point shock to oil price inflation** (ε_{oil} = 10.0 at t = 0). Given the oil price persistence parameter (ρ_{oil} = 0.70), the oil price inflation follows:

$$\pi^{oil}_t = 0.70 \times \pi^{oil}_{t-1} + \varepsilon_{oil,t}$$

This generates a gradually declining oil inflation path: 10.0 pp on impact, 7.0 pp in Q1, 4.9 pp in Q2, declining to below 1 pp by Q8. The cumulative oil price level increase over five years is approximately **33.3 percent**.

Transmission channels to Lesotho

The oil shock transmits to Lesotho through the **differential CPI weight channel**. The Lesotho Phillips curve is:

$$\pi^{LSO}_t = \pi^{ZAF}_t + (\omega^{LSO}_1 - \omega^{ZAF}_1) \times \pi^{oil}_t + (\omega^{LSO}_2 - \omega^{ZAF}_2) \times \pi^{food}_t + \beta_1 y^{LSO}_t + \ldots$$

Since Lesotho has a higher oil weight in its CPI basket (8%) compared to South Africa (5%), the **differential effect is 3 percentage points per unit of oil inflation**. On impact: (0.08 − 0.05) × 10.0 = **0.30 pp** of direct inflation pass-through, with an additional 0.02 pp from the demand feedback (output gap × β_1).

> ***Key insight:*** *Under Lesotho's currency peg to the Rand, the Central Bank of Lesotho (CBL) cannot independently tighten monetary policy in response to the oil-driven inflation. The CBL must track SARB's rate, but since South Africa's Phillips curve in this model does not include direct commodity price terms, SARB does not respond. This creates an* ***asymmetric inflationary burden*** *on Lesotho.*

2. Macroeconomic Impact

2.1 Summary of Results

Variable	Impact (Q0)	Peak	Peak Qtr	Year 1 Avg	Year 2 Avg
Oil Price Inflation (pp)	+10.00	+10.00	0	+6.33	+1.52
Lesotho Inflation (pp)	+0.32	+0.32	0	+0.23	+0.09
Lesotho Output Gap (pp)	+0.09	+0.19	5	+0.14	+0.19
Lesotho Real Interest Rate (pp)	-0.26	-0.26	0	-0.22	-0.19
CBL Interest Rate (pp)	-0.01	-0.18	18	-0.04	-0.12
Real Exchange Rate Gap (pp)	-0.08	-0.15	3	-0.13	-0.14
Reserves Gap (months)	+0.02	+0.37	18	+0.08	+0.23
Risk Premium (pp)	-0.01	-0.18	18	-0.04	-0.12

Table 1: Impulse response summary statistics for a 10% oil price shock

2.2 Impulse Response Functions

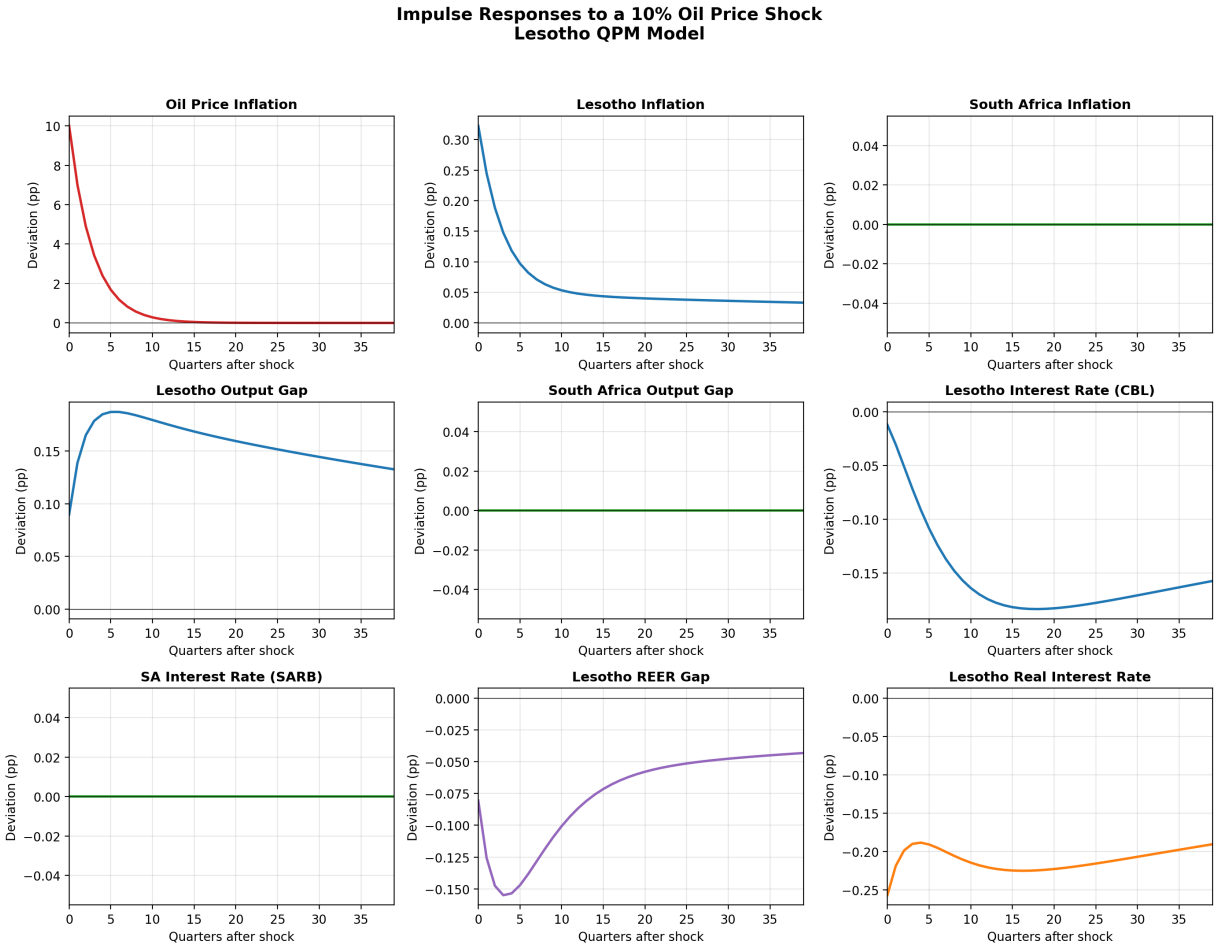

Figure 1: Impulse responses of key macroeconomic variables to a 10% oil price shock. The shock raises Lesotho inflation on impact while lowering the real interest rate, producing a mildly expansionary output effect under the peg.

3. Transmission Mechanism Analysis

3.1 Inflation Channel

The oil shock affects Lesotho inflation through two reinforcing channels:

1. **Direct commodity pass-through:** The differential oil CPI weight ($\omega^{LSO}_1 - \omega^{ZAF}_1 = 0.03$) transmits 0.30 pp of inflation on impact. This is the dominant channel, accounting for approximately 93% of the initial inflation response.
2. **Demand feedback:** The positive output gap ($\beta_1 = 0.25$) adds a further 0.02 pp through the Phillips curve demand channel.

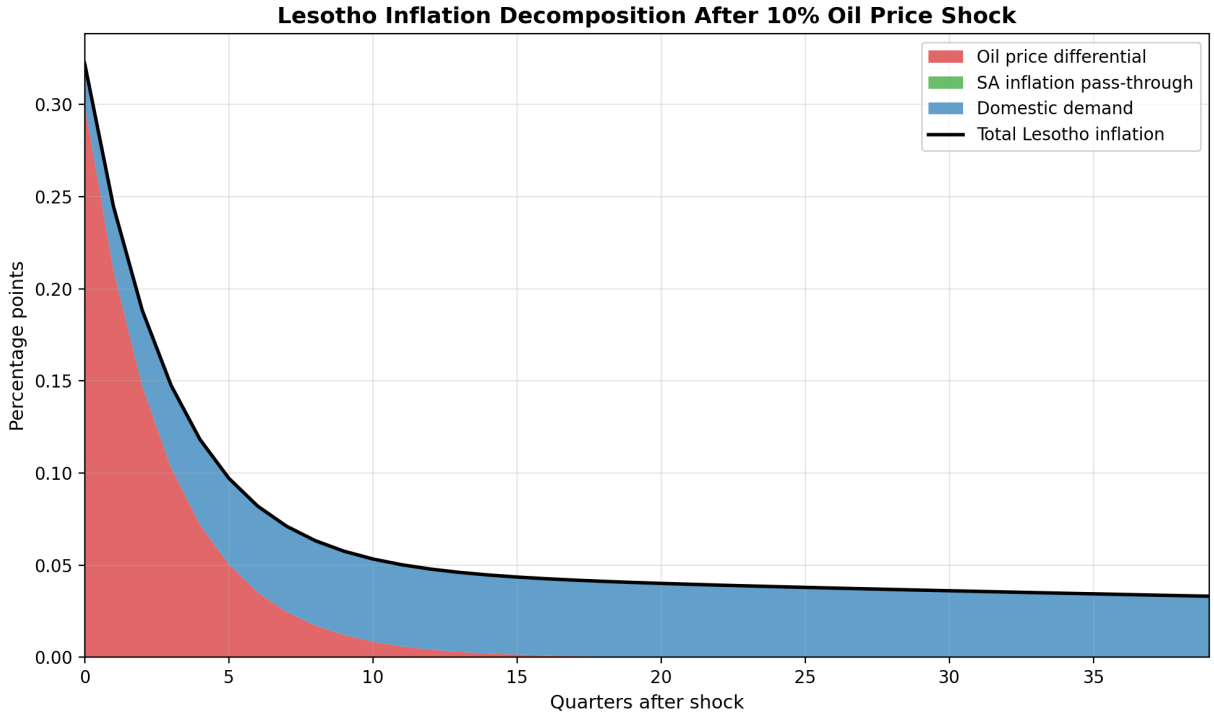

Figure 2: Decomposition of Lesotho inflation response. The oil price differential effect (red) dominates, with a small contribution from demand pressure (blue). SA inflation pass-through (green) is negligible as the SA Phillips curve does not include direct commodity price effects.

3.2 Real Interest Rate and Output Channel

The most notable result is the **expansionary output effect** of the oil shock. This seemingly counterintuitive result follows directly from the peg constraint:

1. Oil shock raises Lesotho inflation by 0.32 pp
2. The CBL cannot independently raise rates (must track SARB)
3. SARB does not respond (oil does not enter SA Phillips curve directly)

4. Real interest rate falls: $r^{LSO} = i^{LSO} - E_t\pi^{LSO}_{t+1}$ drops by 0.26 pp
5. Lower real rate stimulates demand through the IS curve

> ***Policy implication:*** *Under the peg, oil price shocks create a procyclical monetary stance in Lesotho. Higher inflation lowers real rates, stimulating demand when the central bank should be tightening. This highlights the cost of limited monetary autonomy when facing asymmetric supply shocks.*

3.3 External Sector: Reserves and Risk Premium

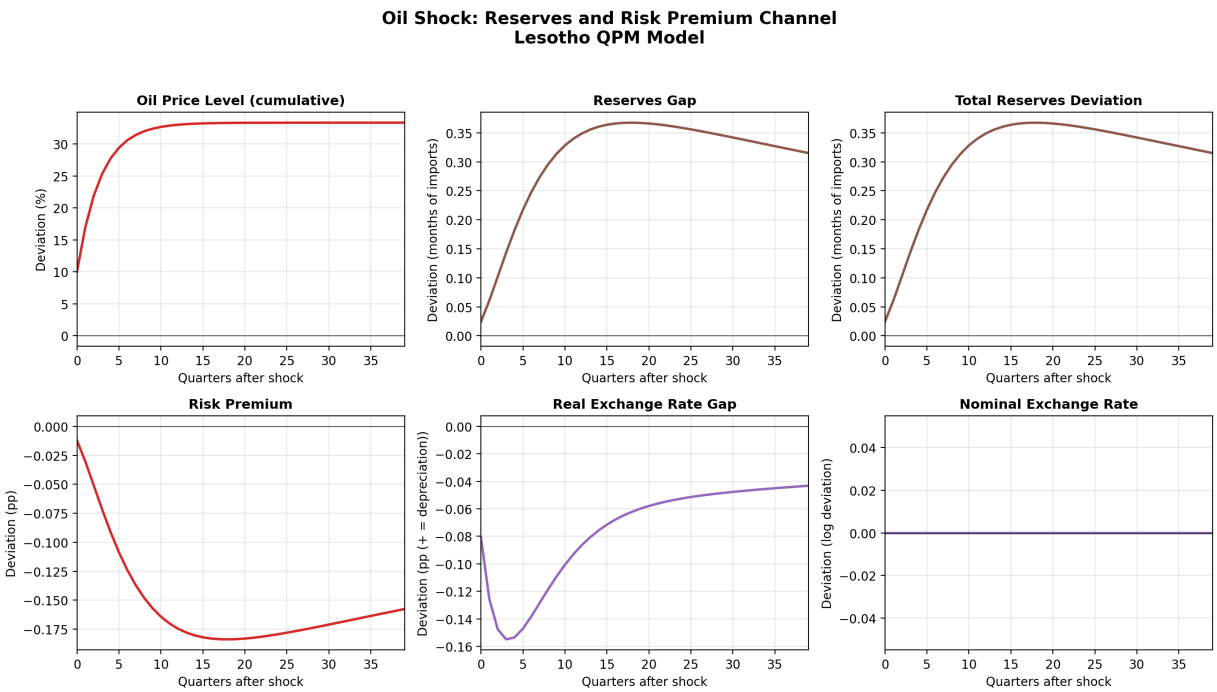

Figure 3: Oil shock transmission through the reserves and risk premium channel. The REER appreciation (lower panel, left) reduces reserve pressure, slightly improving the reserves gap and lowering the risk premium.

The oil shock produces a **modest improvement in reserves** through an indirect channel:

- Higher Lesotho inflation relative to SA causes **real appreciation** (REER gap = -0.08 pp)
- In this model, REER appreciation reduces exchange rate pressure on reserves ($f_2 \times z$), improving the reserves gap
- Higher reserves reduce the risk premium, further easing monetary conditions

This creates a **positive feedback loop** that amplifies the initial expansionary impulse: oil shock → inflation ↑ → real rate ↓ → output ↑ → REER appreciation → reserves ↑ → premium ↓ → output ↑ further.

4. Lesotho vs. South Africa Comparison

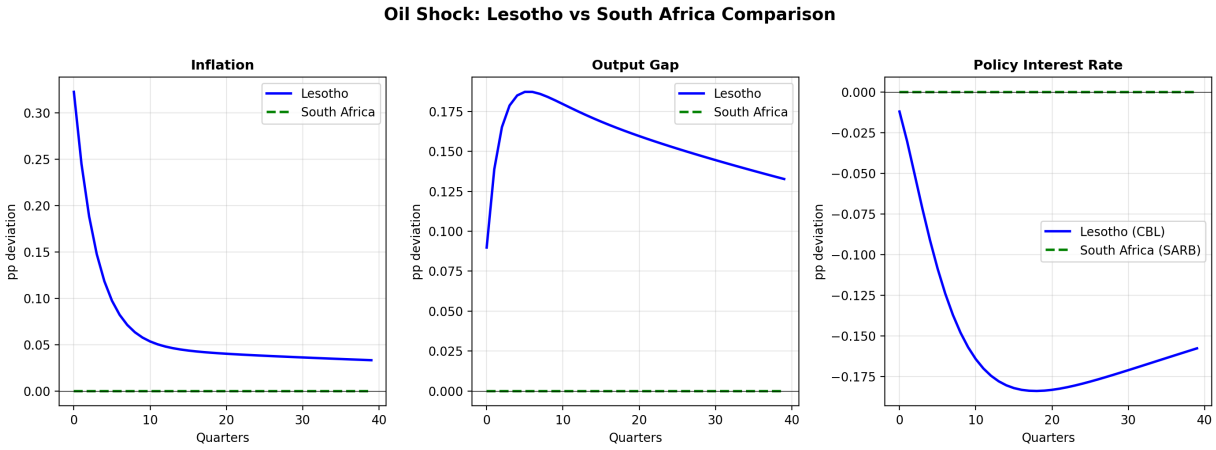

Figure 4: Comparison of Lesotho and South Africa responses. Lesotho bears the full brunt of the inflation shock due to its higher CPI oil weight, while South Africa is insulated. The asymmetric response highlights the cost of the peg when facing commodity-specific shocks.

The comparison reveals the **fundamental asymmetry** of commodity shocks under the peg arrangement:

Variable	Lesotho	South Africa	Implication
Inflation (impact)	+0.32 pp	0.00 pp	Lesotho bears all inflation
Output gap (peak)	+0.19 pp	0.00 pp	Procyclical monetary stance
Policy rate (impact)	-0.01 pp	0.00 pp	No independent response possible
Real rate (impact)	-0.26 pp	0.00 pp	Monetary conditions loosen

Table 2: Asymmetric impact of oil shock on Lesotho vs. South Africa

> ***Structural vulnerability:*** *The model confirms that Lesotho's higher dependence on oil imports (8% of CPI vs. 5% for SA) combined with the loss of monetary autonomy under the peg creates an asymmetric inflation burden. South Africa can absorb oil shocks through its flexible monetary policy framework; Lesotho cannot.*

5. Sensitivity and Robustness

5.1 Shock Size Proportionality

Since the model is linear, all results scale proportionally with the shock size. A 20% oil shock would produce exactly double the inflation and output responses. A 50% shock (similar to the 2022 energy crisis) would generate approximately:

- Inflation: +1.6 pp on impact
- Output gap: +0.94 pp peak expansion
- Real rate: -1.3 pp decline

5.2 Key Parameter Sensitivity

Parameter	Base Value	Role in Oil Shock Transmission
ω^{LSO}_1 (oil CPI weight)	0.08	Direct pass-through; doubling to 0.16 would triple inflation impact
ρ_{oil} (oil persistence)	0.70	Governs duration; higher values prolong inflationary period
β_1 (Phillips curve slope)	0.25	Amplifies demand feedback; steeper curve raises second-round effects
α_4 (monetary conditions)	0.45	Determines output sensitivity to real rate decline

Table 3: Key parameters governing oil shock transmission

6. Policy Implications

6.1 The CBL's Constrained Response

The oil shock analysis reveals a fundamental limitation of Lesotho's monetary arrangement. Under the peg, the CBL faces a **"trapped accommodation" problem**:

- Oil shock raises domestic inflation above SA levels
- The peg requires CBL to track SARB's policy rate
- SARB has no reason to tighten (oil doesn't directly affect SA in this model)
- Result: **real monetary conditions loosen precisely when they should tighten**

6.2 Recommended Policy Responses

Given the monetary policy constraint, the authorities could consider:

1. **Targeted fiscal measures:** Temporary fuel subsidies or tax relief to partially offset the CPI impact (though this has fiscal cost and may drain reserves if sustained).
2. **Macroprudential tightening:** Reserve requirements or loan-to-value limits could cool demand without violating the peg.
3. **Strategic petroleum reserves:** Building oil buffer stocks during periods of low prices to smooth domestic price volatility.
4. **Energy diversification:** Reducing the CPI oil weight through investment in renewable energy and energy efficiency would structurally reduce vulnerability.

6.3 Reserve Adequacy Considerations

While the oil shock modestly *improves* reserves in this simulation (through the REER channel), a sustained oil price increase would eventually raise import costs, potentially reversing this effect. Maintaining reserves above the ARA target of 4.7 months of import cover is essential to preserve the credibility of the peg arrangement during commodity price episodes.

7. Model Limitations and Extensions

- **No direct SA oil channel:** The SA Phillips curve does not include oil prices directly. In reality, SARB would also face higher headline inflation and may tighten, partially mitigating the Lesotho real rate decline.
- **Linear model:** Cannot capture threshold effects (e.g., reserves falling below a critical level triggering nonlinear premium responses).
- **No second-round effects on food:** Oil price increases raise agricultural input costs, which would amplify the shock through Lesotho's high food CPI weight (35%).

- **No terms-of-trade deterioration:** Rising oil import costs worsen Lesotho's current account, which could create additional reserve pressure not captured here.

8. Conclusion

The simulation of a 10% oil price shock through the Lesotho QPM reveals the structural vulnerability of a small, import-dependent economy operating under a currency peg. The key findings are:

1. **Immediate inflation impact:** 0.32 pp rise in Lesotho CPI inflation on impact, driven by the differential oil weight in the consumption basket.
2. **Procyclical monetary conditions:** Unable to tighten independently, Lesotho experiences a 0.26 pp real interest rate decline, producing a counterintuitive output expansion of up to 0.19 pp.
3. **Persistent effects:** With oil price persistence of 0.70, inflation remains elevated for 8–10 quarters, averaging 0.23 pp in year 1 and 0.09 pp in year 2.
4. **Asymmetric burden:** Lesotho bears the full inflationary cost while South Africa is insulated, highlighting the price Lesotho pays for monetary policy subordination under the CMA arrangement.

These results underscore the importance of **non-monetary policy instruments**—fiscal buffers, macroprudential tools, and structural energy reforms—for managing commodity price shocks in Lesotho.

Technical note: Model solved using iterative perfect-foresight method (converged in 33 iterations). All results are percentage point deviations from steady state. Based on the Lesotho QPM v2 (recalibrated), implemented from IMF Country Report No. 2023/269.